

Day 106 Project

Modeling Stock Returns with ARIMA + GARCH

1 Introduction

This project combines ARIMA and GARCH models to jointly forecast returns and volatility.

The goal is to model both the conditional mean and conditional variance of financial time series.

2 Model Framework

We decompose returns as:

$$r_t = \mu_t + \epsilon_t$$

where:

- μ_t is modeled using ARIMA
- $\epsilon_t \sim N(0, \sigma_t^2)$

3 ARIMA for Mean

$$\mu_t = c + \sum_{i=1}^p \phi_i r_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j}$$

This captures autocorrelation in returns.

4 GARCH for Volatility

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

This captures volatility clustering.

5 Estimation Procedure

1. Fit ARIMA model to returns
2. Extract residuals ϵ_t
3. Fit GARCH model to residuals
4. Forecast future mean and volatility

6 Forecasting

- ARIMA provides return forecasts
- GARCH provides volatility forecasts

7 Financial Interpretation

- Expected return $\rightarrow$ trading signal
- Volatility $\rightarrow$ risk measure

8 Applications

- portfolio optimization
- risk management (VaR)
- algorithmic trading

9 Conclusion

Combining ARIMA and GARCH provides a complete framework for modeling financial time series, capturing both signal and risk.